

Machine Learning and AI: Study Notes

What is Machine Learning?

Machine Learning (ML) is a subfield of Artificial Intelligence that enables computers to learn patterns from data without being explicitly programmed. Instead of writing fixed rules, an ML model is trained on examples and gradually improves its performance on a task, such as classification or prediction, by minimizing an error metric during training.

Types of Machine Learning

There are three main types of machine learning. Supervised learning uses labeled data, where each input has a known output, to train models such as regression and classification algorithms.

Unsupervised learning works with unlabeled data to discover hidden patterns, such as clustering and dimensionality reduction. Reinforcement learning trains an agent to make sequential decisions by rewarding desirable actions and penalizing undesirable ones.

Neural Networks

A neural network is a computational model inspired by the human brain, composed of layers of interconnected nodes called neurons. Each connection has a weight that is adjusted during training through backpropagation, which uses gradient descent to minimize a loss function. Deep learning refers to neural networks with many hidden layers, which allow the model to learn increasingly abstract representations of the input data.

Overfitting and Regularization

Overfitting occurs when a model learns the training data too well, including its noise, and performs poorly on new, unseen data. Common techniques to reduce overfitting include regularization methods such as L1 and L2 penalties, dropout layers in neural networks, early stopping during training, and increasing the size or diversity of the training data.

What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation is a technique that combines a retrieval system with a generative language model. Instead of relying only on knowledge stored in the model's parameters, RAG retrieves relevant text chunks from an external document collection using vector similarity search, and feeds those chunks as context to the language model so it can generate an answer grounded in the retrieved evidence. This reduces hallucination and allows question answering over private or domain-specific documents.

Vector Embeddings

A vector embedding is a numerical representation of text that captures its semantic meaning in a high-dimensional space. Sentences with similar meaning are mapped to nearby points in this space. Embedding models, often built on transformer architectures, convert text chunks and user queries into embeddings so that similarity search, typically using cosine similarity, can identify the most relevant

chunks for a query.

Evaluation Metrics for Classification

Common evaluation metrics for classification models include accuracy, precision, recall, and F1-score. Accuracy measures the overall proportion of correct predictions. Precision measures how many predicted positives are actually correct, while recall measures how many actual positives were correctly identified. The F1-score is the harmonic mean of precision and recall, and is especially useful when classes are imbalanced.